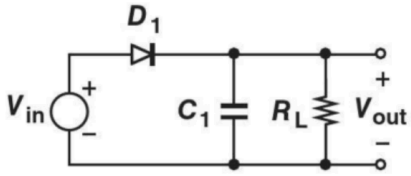
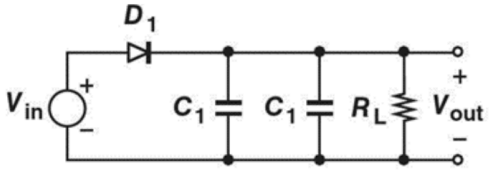
Midterm

5 Problems (20 points each, total 100 points)

1. The half-wave rectifier shown below drives a 100-Ω load through a diode with $V_{D,on} = 0.6\,\text{V}$ and a 500-$\mu$F smoothing capacitor as shown in (a). The input is a 60-Hz sinusoidal voltage source with a peak voltage of 3.5 V. If the circuit is replaced by (b), Calculate the ripple amplitude respectively (20 points)



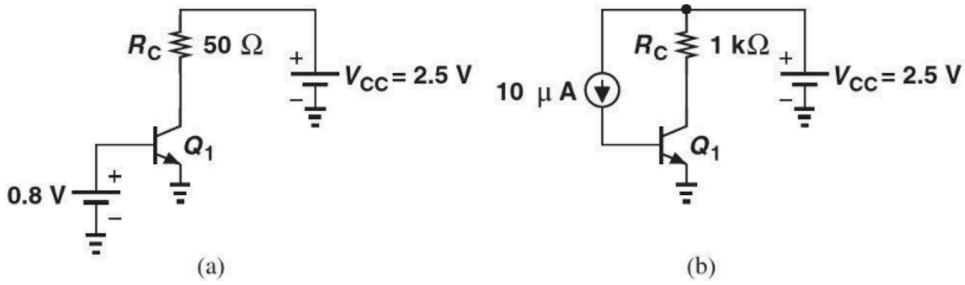

(a)




(b)

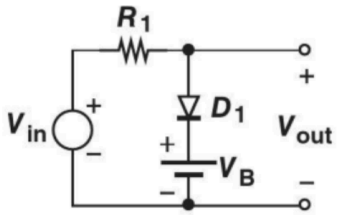
2. A junction employs $N_D = 4 \times 10^{16}\ \text{cm}^{-3}$ and $N_A = 3 \times 10^{15}\text{cm}^{-3}$ Calculate the built-in potential at (a) T = 300 K and (b) 350 K. (20 points)

3. Determine the operating point and the small-signal model of Q for the circuits shown in (a) and (b). Assume $I_S = 2\text{x}10^{-16}$ A, $\beta = 100$, and $V_A = \infty$. (20points)

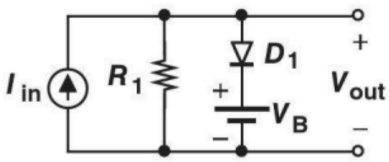


4. A bipolar transistor carries a collector current of 1mA with $V_{CE} = 2V$. Determine the required base-emitter voltage if (a) $V_A = \infty$ or (b) $V_A = 20V$. Assume $I_S = 3 \times 10^{-16}$ A.

5. In the circuits of (a) and (b), plot the current flowing through $R_1$ as a function of $V_{in}$ *or* $I_{in}$. Assume a constant-voltage diode model with $V_{Don} = 0.8V$ and $V_B = 1V$ (20points)




(a)




(b)